Airbnb Price Prediction — Summary Report

1. Objective

The objective of this project was to build a Linear Regression model to predict Airbnb listing prices in New York City using features such as location, room type, minimum nights, and availability.

2. Model Performance

The model was evaluated on a test dataset containing 20% of the data that was not used during training.

R-squared (R^2) Score: 0.467

The model explains approximately **46.7% of the variation in Airbnb listing prices** based on the features provided. The remaining 53.3% of the variation is not explained by the model. This suggests that other factors, such as amenities, property quality, and demand, may also influence prices.

Mean Absolute Error (MAE): $45.70

On average, the model's predicted price differs from the actual listing price by approximately **$45.70 per listing**. For example, if an actual listing price is $150, a prediction might be higher or lower by around $45.70 on average. This is an average error, not a guarantee for every listing.

Mean Squared Error (MSE): 3,914.93

MSE measures prediction errors after squaring them, which gives greater weight to large errors. Its unit is dollars squared, so it is less intuitive for business interpretation than MAE.

3. Key Findings

The model provides a useful baseline for estimating Airbnb prices, but its R^2 score indicates that its explanatory power is limited. Predictions may be less accurate for listings with unusual or high prices. The actual-versus-predicted scatter plot and residual plot can help identify these errors and patterns.

4. Conclusion

The Linear Regression model offers an initial estimate of Airbnb listing prices, with an average absolute error of approximately $45.70 and an R^2 score of 0.467. Before using it to support business pricing decisions, the model could be improved by including more relevant features, exploring nonlinear relationships, and evaluating alternative models.